Project Design Phase

Proposed Solution Template

Date	07 July 2026
Team ID	SWTID-2026-2247
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Proposed Solution Template:

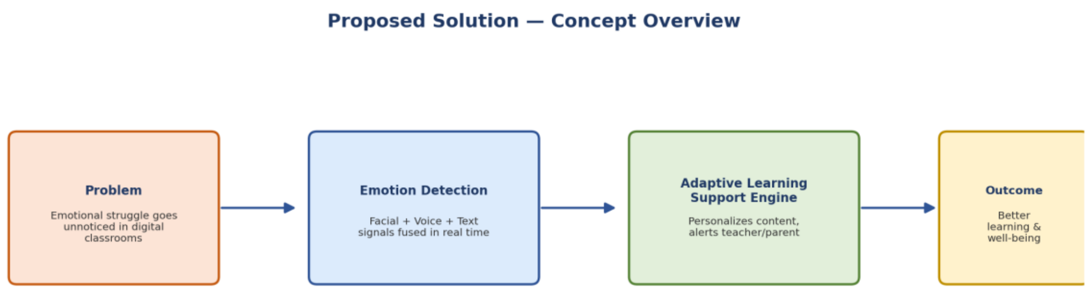
Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	**Problem Statement (Problem to be solved)**	● Students in online/hybrid classrooms often experience confusion, boredom, stress, or disengagement that goes unnoticed until it shows up in poor grades or dropout. ● Existing e-learning platforms adapt only to quiz scores or time spent, not to how the learner actually feels, leaving struggling students without timely support.
2.	**Idea / Solution description**	● An AI-powered Emotion Detection and Learning Support Engine that captures student emotional signals in real time through webcam (facial expression), microphone (voice tone), and chat/quiz text (sentiment). ● Classifies the emotional state (confused, bored, stressed, engaged) and feeds it into an adaptive learning engine that personalizes content difficulty, suggests breaks, and alerts teachers/parents instantly. ● Integrates directly into existing LMS and video-conferencing tools rather than requiring a separate app.
3.	**Novelty / Uniqueness**	● Multimodal real-time emotion fusion (facial + voice + text) instead of relying on a single signal. ● Direct plug-in integration into existing classroom tools (LMS, Zoom, Teams) rather than a standalone platform. ● Closed-loop adaptive learning: emotional state directly drives content/pace changes, not just analytics dashboards. ● Privacy-first design with encrypted data handling and edge-processing options suited for minors.
4.	**Social Impact / Customer Satisfaction**	● Reduces student dropout and learning anxiety by catching struggles early. ● Empowers teachers with actionable, real-time insight instead of guesswork. ● Gives parents visibility into their child's emotional learning journey. ● Promotes inclusive education by supporting students reluctant to ask for help, improving overall well-being.
5.	**Business Model (Revenue Model)**	● SaaS subscription for schools/institutions with per-student/per-classroom tiered licensing.

		• Freemium model for individual tutors/small coaching centers with a premium analytics add-on. • B2B2C partnerships with LMS providers (Google Classroom, Moodle) for bundled integration. • Aggregated, anonymized insights/reporting offered as a premium add-on for institutions.
6.	**Scalability of the Solution**	• Cloud-native microservices architecture (Docker + Kubernetes) scales from a single classroom to district/national level. • Multi-language support for facial/voice/text emotion models enables expansion across regions. • API-first design allows integration with any LMS/video platform for horizontal market growth. • Modular AI models extend to new age groups, corporate training, and government literacy programs.

Solution Diagrams:

1. Solution Concept Overview



2. Scalability Path